AIR FRYERS UNDER THE MICROSCOPE: DEMAND, DIFFERENTIATION, AND MARKET POWER

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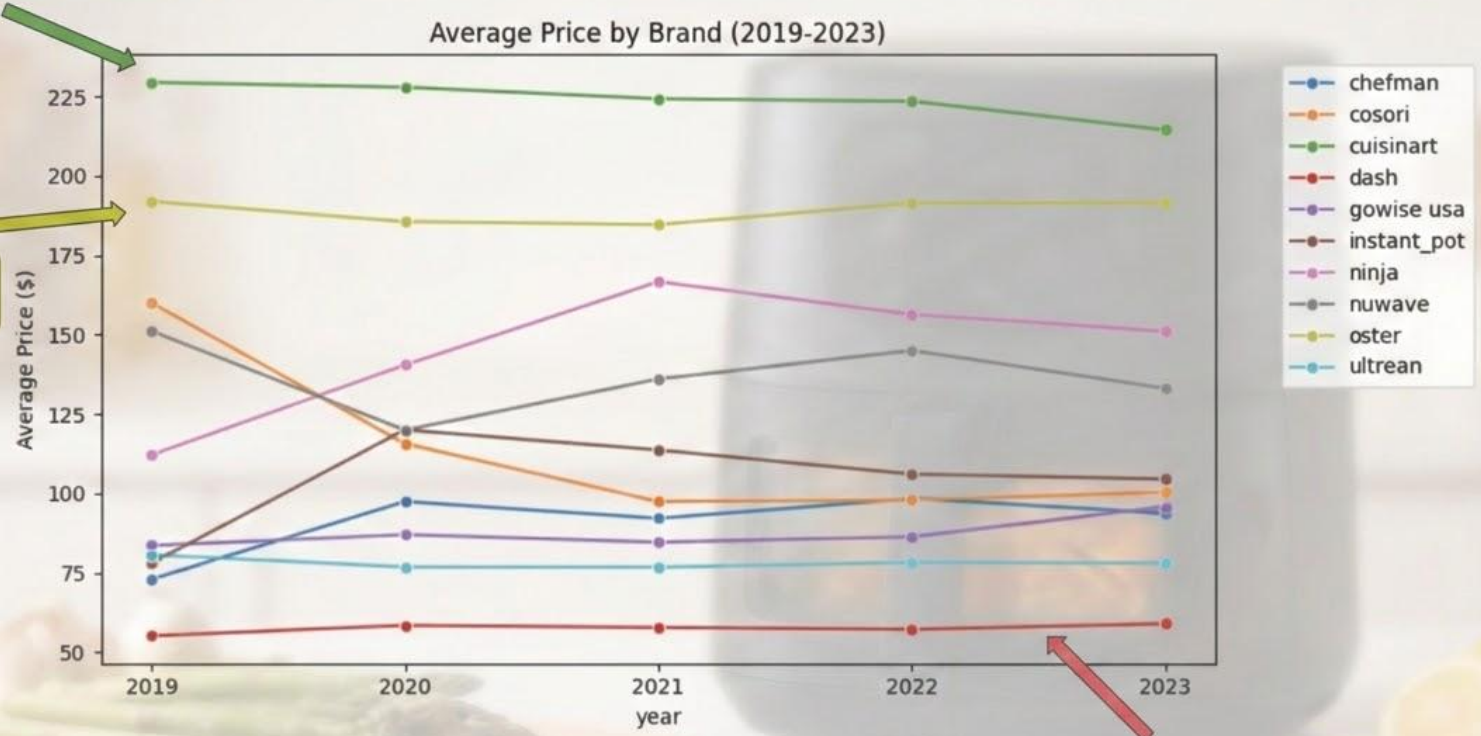
Market Overview: Air Fryer (2019-2023)

The air fryer market is a really competitive and fast -changing part of the small kitchen appliance industry. Between 2019 and 2023, it went from being kind of a niche product to something a lot of people have in their kitchens, and different brands have been competing pretty heavily for market share. The market is basically split into tiers. Cuisinart is more on the premium side with multi functional designs, while Dash focuses on cheaper, smaller options. In the middle, brands like Ninja and Instant Pot have done really well because they already have strong brand recognition. Most air fryers are still pretty compact, but companies are starting to add features like viewing windows and dual baskets to stand out and charge a bit more, even though people are still pretty price conscious.

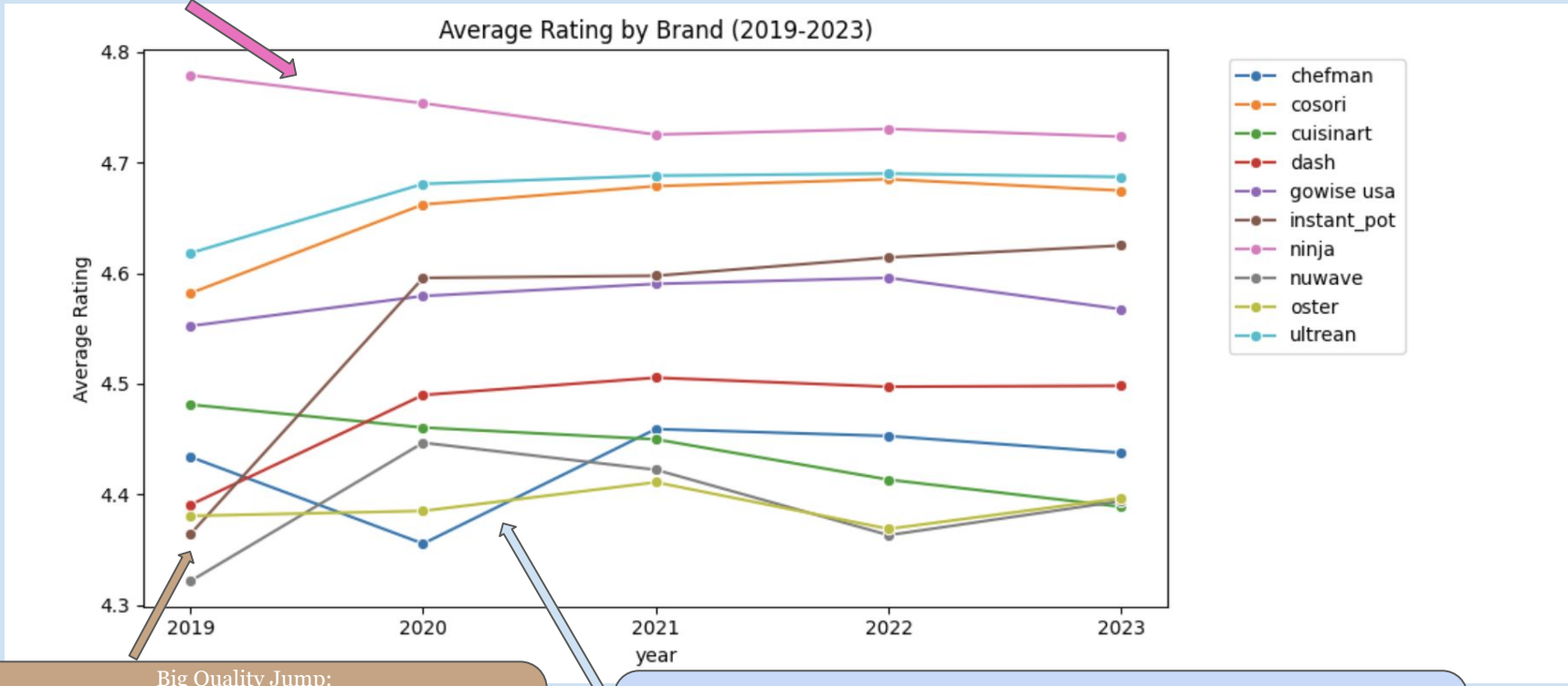
Cuisinart air fryers have the highest average price



Oster are consistently high-priced

Dash air fryers have the highest average price

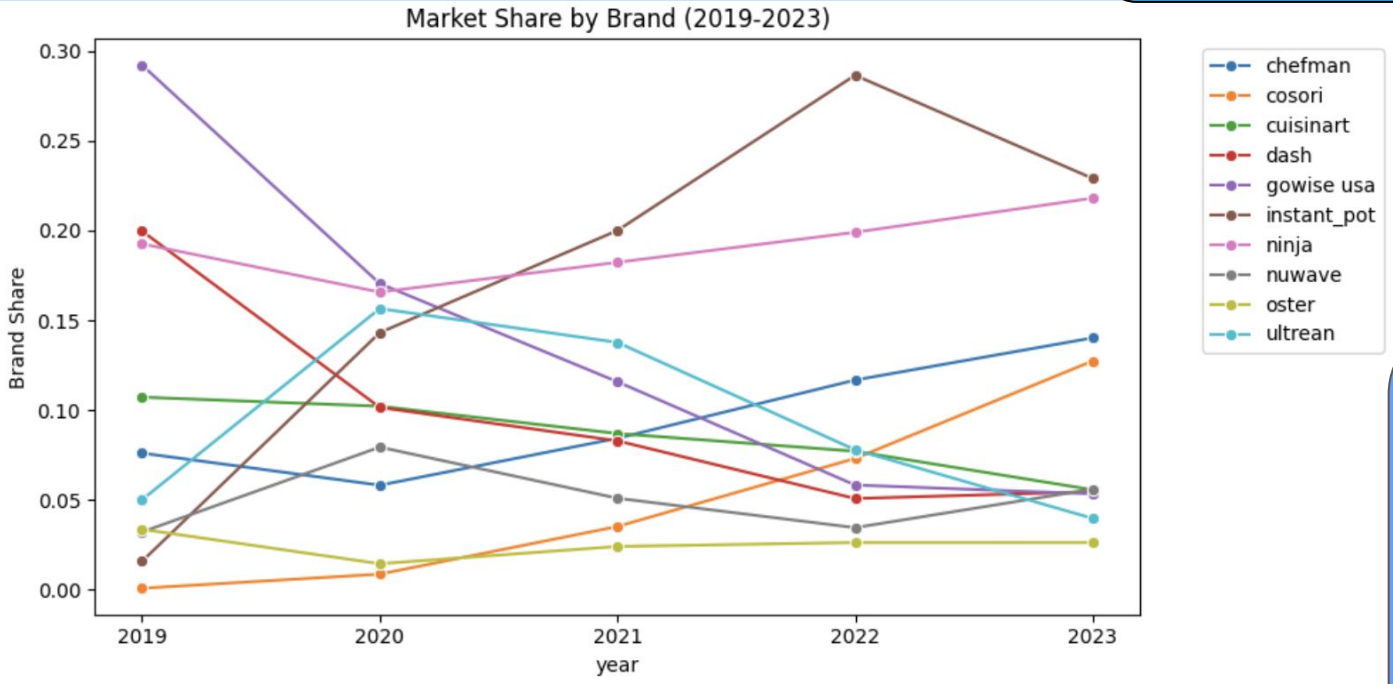
Ninja air fryers have the highest average rating!



Big Quality Jump:
Instant Pot's rating jumps from around 4.35 to almost 4.6 between 2019 and 2020, showing they likely improved their products or released higher-quality models, which helped boost how people viewed the brand.

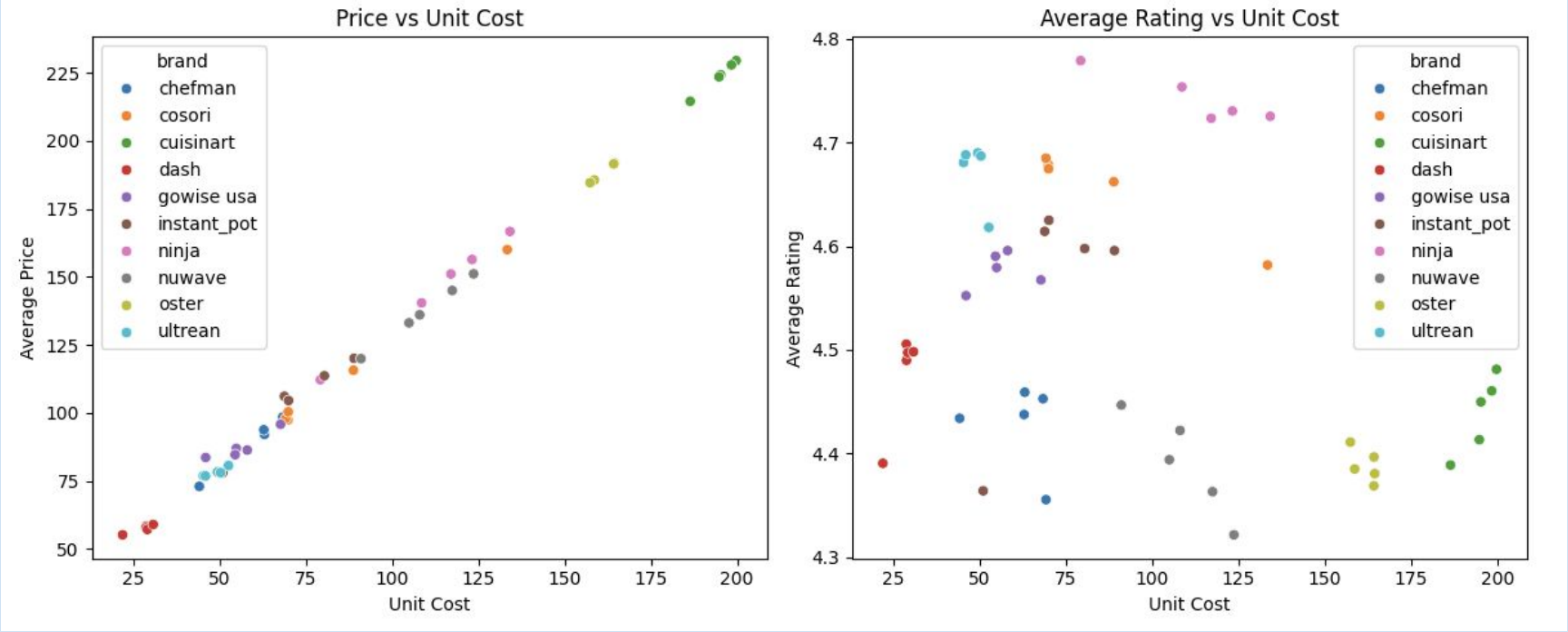
Ultrean, usually a solid budget/mid-tier brand,saw its rating slip, not because it got worse, but because the market improved. A 4.4 was strong in 2019, but by 2020, with Instant Pot closer to 4.6, it looked average. This shows that in this market, standing still is basically falling behind.

Even though brands differ a lot in price, ratings are almost identical across the board. That tells us consumers aren't really choosing based on satisfaction differences. At the same time, market share shifts a lot over time, no brand dominates consistently. So this is a very competitive market where brand strength and positioning matter more than just pricing or product quality.



Upward Momentum: Ninja and Instant Pot have clearly grown since 2019. Cosori shows a very consistent, steep climb from nearly zero.

Downward Trend: GoWise USA started as the market leader in 2019 but has seen a massive decline. Dash and Cuisinart have also lost significant share over the five-year period.



There is a strong positive relationship. Models with higher unit costs are sold at higher prices. This suggests consistent markup strategies across brands, with only small differences.

Ratings stay within a narrow range, about 4.3 to 4.8, and do not clearly increase with cost. Higher priced models do not necessarily get better reviews.

Cost strongly affects price, but not customer satisfaction. Mid range models can perform just as well as premium ones.

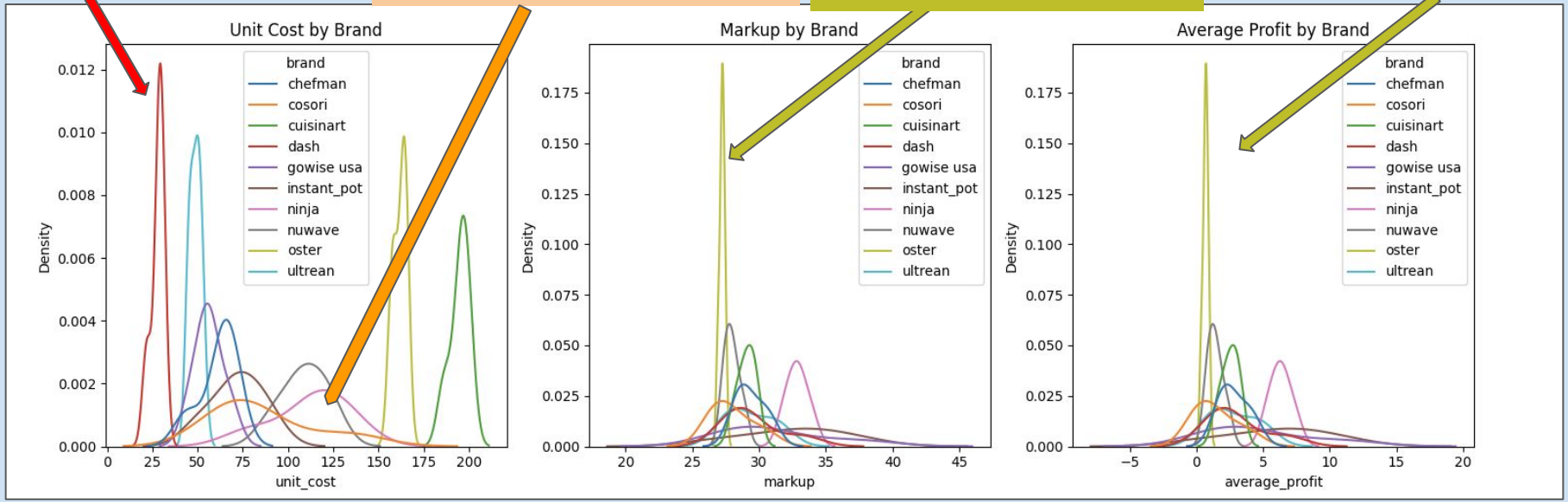
Market Share Over Time

Dash has the highest unit cost which so they're probably using more expensive materials or designs.

Cosori maintains one of the lowest production cost in the group, allowing them to dominate the budget tier.

Oster maintains the highest markup in the category. This suggests they have the best balance between a high selling price and low production costs.

Oster leads the market in total value, showing that its higher markups are effectively translating into the highest overall earnings.



- Cost lines aren't perfectly flat, they shift as the market changes
- These changes reflect the behind the scenes competition between brands
- When costs drop, it usually means a brand is becoming more efficient and saving money
- When costs rise, brands are likely adding better components or improving products to stay competitive
- These movements help explain why some brands are more profitable than others year to year

- This graph shows the "true value" of each brand. Oster earns more profit per unit than any other competitor.
- The highest markup brand (Oster) is not the same as the highest market share brand (Ninja), showing two different strategies: profit vs. volume.
- The "profit leader" changes over time depending on which brand manages costs most efficiently.

- Oster combines high margins with strong enough market presence to lead in total revenue
- Profit changes more sharply than price, meaning small shifts in cost or demand can quickly affect earnings
- Some brands focus on premium profits (Oster), while others focus on high volume sales, here, the premium strategy performs best

The Demand Model Equation

$$\hat{s}_{bt}(p_{bt}) = \hat{\beta}_{price} s_{bt}(1 - s_{bt}).$$

The Method:
To analyze the market, we used a multinomial logistic regression, but simplified it by running a linear regression on the log of market share.

What this means:
This formula basically helps us figure out what actually influences a customer's choice.

Breaking it down:

- **Predicting Share (log share):** We're looking at what drives sales for each brand over time.
- **Price:** Shows how sensitive customers are to price changes.
- **Quality (ratings):** Tells us if better reviews actually lead to more sales.
- **Features:** Shows the added value of things like viewing windows or compact designs.
- **Brand Identity:** Captures the effect of brand reputation (like Ninja vs. lesser-known brands).

Result:
The model has an R^2 of 0.76, meaning it explains about 76% of the market's behavior, which makes it pretty reliable for our pricing and product decisions.

How Price Affects Demand

The Key Number: The estimated price coefficient is $\hat{\beta}$_price = −0.0377

What does the negative sign mean?

- A higher price → lower market share. This is the law of demand confirmed in the data.
- Specifically, a $1 increase in average price is associated with a ~3.8% decrease in log market share, holding brand, year, and product features constant.
- If the coefficient were positive, it would mean consumers buy more as prices rise. This makes no economic sense and would signal a broken model. The negative sign validates the entire analysis.

Why does it matter downstream?

- Since it's shared across all brands and years, it gives a clean, apples-to-apples comparison of cost and profitability across the market.

Important Product Features

Compact design dominates the market (98%)

Almost all air fryers are compact, making this a standard feature across brands. It is now a basic expectation rather than a differentiator, so it does not strongly justify higher pricing.

Oven-style design is widely adopted (56.3%)

More than half of models use an oven-style layout instead of the traditional basket design. This shows a shift in design preference, but it is no longer unique enough to stand out.

Key takeaway

Both compact size and oven-style design are common across the industry. Since many brands already offer these features, they provide limited competitive advantage on their own.

Here’s an example of a typical air fryer. Compact and oven-style, just like most models on the market.

Unit Costs, Markups, and Profitability

Brand	Unit Cost	Markup	Avg Profit (share-weighted)
Cuisinart	194.90	29.05	2.50
Oster	161.80	27.23	0.68
Ninja	112.49	32.85	6.31
Instant Pot	71.90	32.56	6.01
Dash	27.94	29.54	2.99

Key Takeaways

- Markups are surprisingly similar across brands (~$27-$33), despite wildly different unit costs
- Ninja and Instant Pot dominate profitability, combining strong markups with large market share
- Oster and Cosori underperform, with decent markups but too small a share to matter
- Higher unit costs do not guarantee higher profits (see Cuisinart vs. Ninja)

Conclusion: Market Positions

- Leaders: Ninja, Instant Pot
  - High market share + strong brand effects
  - Not the most expensive, but highly preferred

➔ Strong branding and broad appeal

- Premium: Cuisinart, Oster
  - Highest prices and highest unit costs
  - Does not translate into dominant market share

➔ Premium pricing ≠ market leadership

- Budget Tier:
  - Lowest prices and costs
  - Limited market share and profit

➔ Competes on price, but lacks scale

Overall, brand strength and positioning are the key drivers of success, while product features and pricing alone are not enough to win in the airfryer market.